Repeated Transaction Detection

Program explanation and documentation

1. Aim

To develop a Python program that detects potentially suspicious transactions by checking for repeated transaction IDs and transaction amounts that exceed a specified threshold.

2. Description

The program accepts the number of transactions and a suspicious-amount threshold from the user. It then reads each transaction ID and amount. A set stores transaction IDs already encountered, allowing the program to identify repeated IDs efficiently. A list stores every transaction flagged by at least one of the two checks.

3. Input

Input	Purpose
Number of transactions (n)	Specifies how many transactions will be checked.
Suspicious amount threshold	Amount limit used to flag high-value transactions.
Transaction ID	Unique identifier used to detect repeated transactions.
Transaction amount	Amount checked against the threshold.

4. Processing Logic / Algorithm

1. Start the program.

2. Read the number of transactions and the suspicious amount threshold.

3. Create an empty set named **transaction_ids** and an empty list named **suspicious**.

4. Repeat the following steps for each transaction:

5. Read the transaction ID and amount.

6. If the ID is already in the set, mark the transaction as a duplicate. Otherwise, add the ID to the set.

7. If the amount is greater than the threshold, mark a threshold violation.

8. If either condition is true, append the transaction details and both flags to the suspicious list.

9. After all transactions are processed, display the suspicious transactions. If the list is empty, display that no suspicious transactions were found.

10. Stop.

5. Output

The program displays each flagged transaction's ID and amount. It also displays the applicable reason: **Duplicate Transaction**, **Amount Exceeds Threshold**, or both. If no transaction is flagged, it displays “No suspicious transactions found.”

6. Important Conditions

• A transaction is considered a duplicate when its ID has appeared in an earlier transaction.
• An amount is flagged only when it is strictly greater than the threshold; an amount equal to the threshold is not flagged.
• A transaction can be flagged for one reason or for both reasons.
• The first occurrence of an ID is not marked as a duplicate, but a later occurrence is.

7. Conclusion

The program provides a simple way to identify repeated transaction IDs and unusually high transaction amounts. Using a set for ID checking and a list for storing flagged records makes the logic straightforward and efficient.

Note: This is a basic rule-based screening example. It flags records for review; it does not prove that a transaction is fraudulent.